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***GE Medical Systems***

# **Technical Publications**

**Direction 2100624**

**Revision 1**

## **Signa<sup>®</sup> Advantage<sup>™</sup> 1.0T Quadrature T/L Spine Coil with Positioner**

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**Operating Documentation**

# DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, nature, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call Traffic and Transportation, Milwaukee, WI (414) 785-5052/8\*323-5052 immediately after damage is found. At this time be ready to describe name of carrier, delivery date, consignee name, freight or express bill number, item damaged, and extent of damage.

Complete instructions regarding claim procedures are found in section "S" of the Policy & Procedure Bulletins.

3/12/92



**GE MEDICAL SYSTEMS**

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## LIST OF EFFECTIVE PAGES

| <u>REVISION</u> | <u>DATE</u> | <u>PRIMARY REASON FOR CHANGE</u> |
|-----------------|-------------|----------------------------------|
| REV 0           | 11/15/93    | Initial Release                  |
| REV 1           | 6/18/96     | Add 60cm values to Table 2-2.    |

| <u>PAGE</u>   | <u>REV</u> | <u>PAGE</u> | <u>REV</u> | <u>PAGE</u> | <u>REV</u> | <u>PAGE</u> | <u>REV</u> |
|---------------|------------|-------------|------------|-------------|------------|-------------|------------|
| Title Page    | 1          |             |            |             |            |             |            |
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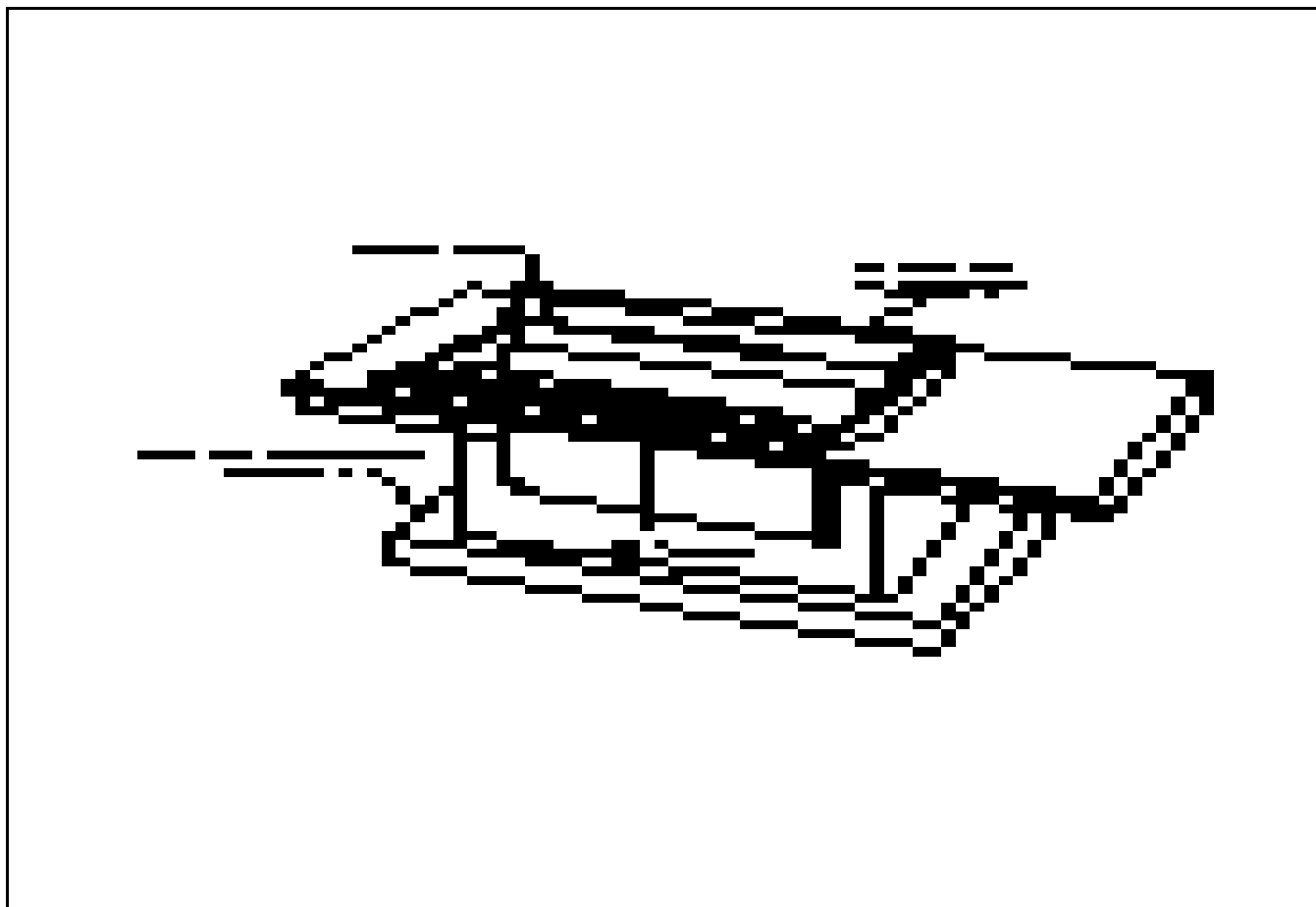
## SECTION 1 - INTRODUCTION

### 1-1 THE QUAD T/L SPINE SURFACE COIL

The Quad T/L Spine Coil is a rigid RF receive-only coil that is designed to image the thoracic/lumbar area or similar region. The coil is optimized for a ROI covering 30 cm along the Z axis, 15 cm along the X axis, and an effective depth of 12 cm above the coil along the Y axis. The coil is tuned to the 1.0T MR frequency, 42.68 MHz.

The Quad T/L Spine Coil can be easily positioned beneath a patient in the supine position. Positioning the coil is performed by sliding a handle on the left anterior side of the coil. A retractable carrying handle is located on the side opposite the Positioning Scale. See ILLUSTRATION 1-1.

The Positioning Scale is graduated in centimeters and is 82 cm long with a "Zero" established at the travel midpoint. The Positioning Handle can move approximately 51 cm total travel length.



QUAD T/LSPINE COIL  
ILLUSTRATION 1-1

**1-3 COMPATIBILITY**

The Quad T/L Spine Surface Coil is compatible with the following hardware configurations:

- GE SIGNA ADVANTAGE™ 1.0T Systems

**1-4 RELATED DOCUMENTS**

- Direction 15400 - Signa Advantage 1.5T, 1.0T, and 0.5T Systems
- Direction 15404 - Signa Advantage 1.5T, 1.0T, and 0.5T Renewal Parts.

**1-5 ORGANIZATION OF THIS DOCUMENT**

This document is divided into the following sections:

Section 1, INTRODUCTION, describes what the Quad T/L Spine Surface Coil is and where the Quad T/L Spine Surface Coil can be used.

Section 2, SET UP AND CALIBRATION, describes installation procedures.

Section 3, FUNCTIONAL CHECKS, describes normal power-up sequence.

Section 4, REPLACEMENT/MAINTENANCE, describes field maintenance procedures

Section 5, RENEWAL PARTS, lists field replaceable service kits and parts.

SNR Data Table

Coil Return Form

TLT Table

**1-6 ENVIRONMENTAL REQUIREMENTS**

Operate and store the Quad T/L Spine Surface Coil in the Scanner Room.



## SECTION 2 - SET UP AND CALIBRATION

### 2-1 CHECKING THE SHIPPING LIST

Table 2-1 lists the M1885AM Signa Quad T/L Spine Coil parts. Check that all parts have been shipped.

TABLE 2-1  
COIL PARTS

| QTY. | ITEM                                | PART NO.    |
|------|-------------------------------------|-------------|
| 1    | Quad T/L Spine Coil with Positioner | 46-328252G1 |
| 1    | Comfort Pad                         | N/A         |
| 1    | Service Manual                      | 2100624     |

### 2-2 INSTALLING THE QUAD T/L SPINE COIL

At the console, install a new soft key. This key will be used by the operator to select Quad T/L Spine imaging. Name the key "**quad t/l**".

Refer to the Signa Advantage 1.5T, 1.0T, & 0.5T System Manual for information on installing soft keys (use Coil default values in Table 2-2)

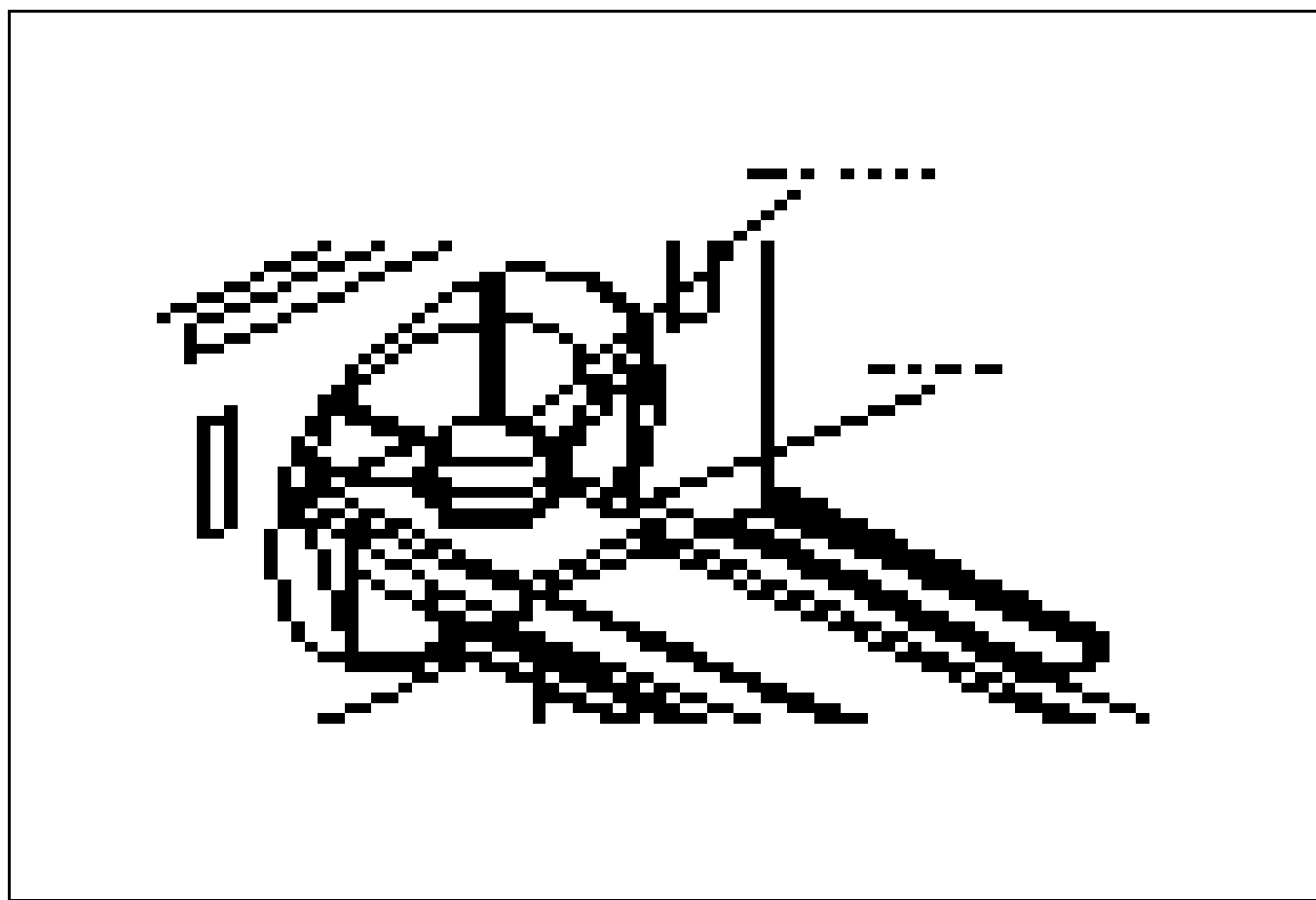
TABLE 2-2  
COIL VALUES

| SYSTEM | COIL NAME | COIL TYPE | EXTREMITY COIL | CABLE LOSS | COIL LOSS                    | RECON SCALE FACTOR                    | XMIT COIL TYPE | XMIT ATTN | MULTI-COIL. |
|--------|-----------|-----------|----------------|------------|------------------------------|---------------------------------------|----------------|-----------|-------------|
| 1.0T   | quad t/l  | surface   | no             | 1.24       | 0.103 - 55cm<br>0.549 - 60cm | Head Coil Recon<br>Scale Fctr. x 1.00 | Quad           | 0         | no          |

### 2-3 DETERMINING SYSTEM POLARITY

Before any quadrature coil can be used on the Signa Advantage™ 1.0T System, the system polarity must be determined and the user coil adapted to reflect the same polarity.

1. Select **[New Exam]** to enable the patient table **LANDMARK** function. Locate the 100 mm phantom on the patient table, see ILLUSTRATION 2-1.
2. Center the 100 mm TLT phantom; X, Y, Z using the positioning lights, **LANDMARK** and advance to isocenter. Set up a scan according to the parameters shown in TABLE 2-3. At the **PREVIEW SCREEN**, press **Scan Ops**. **DO NOT SCAN AT THIS TIME.**
3. Press **[Modify Cvs]**. Type **cfxfull** in the column under **CV Name**.
4. Observe the Current Value column across from cfxfull. If there is a (-) sign in front of the number, the system polarity is **NEGATIVE**. If no sign appears in front of the number, the system polarity is **POSITIVE**.
5. Press **[Enter]** to default to the current value.



PHANTOM PLACEMENT

ILLUSTRATION 2-1

**2-4 CHECKING COIL POLARITY**

Ensure that the SMA connector is connected to the same polarity fitting as the system polarity. If not, remove the access cover shown in ILLUSTRATION 2-2 by removing the screws. Slide the Coil Positioning Handle until the balun board is visible in the access area. Remove the SMA connector from the fitting on the balun board, and connect it to the other fitting.

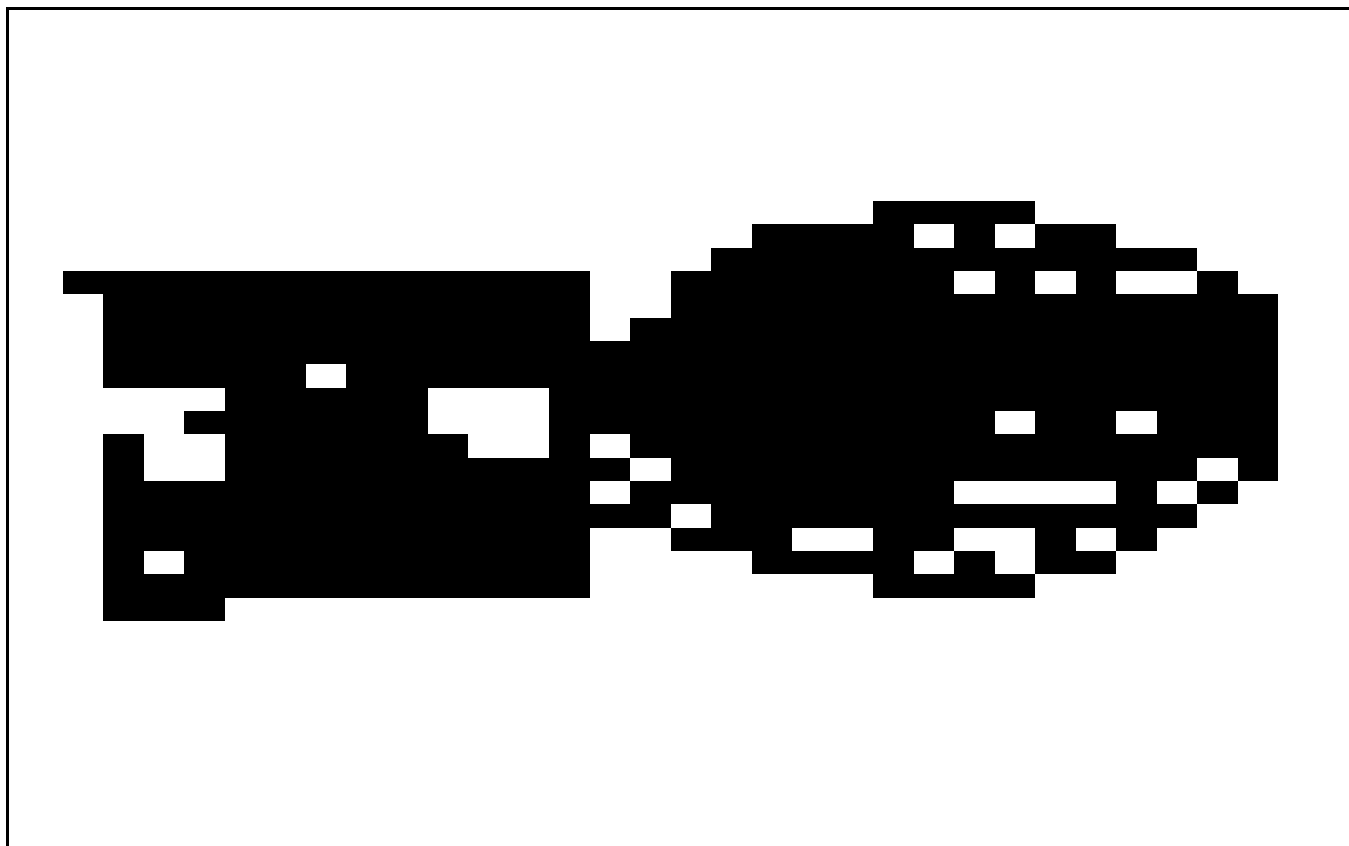
**CHANGING COIL POLARITY**

ILLUSTRATION 2-2

**2-5 FUNCTIONAL CHECKS**

1. Perform a Body coil scanner performance verification. Refer to Section 3-1, SIGNA ADVANTAGE 1.0T SCANNER VERIFICATION.
2. Perform a Quad T/L Spine Coil performance verification. Refer to Section 3-2, QUAD T/L SPINE COIL IMAGING PERFORMANCE VERIFICATION.
3. Perform an SNR Image analysis. Refer to Section 3-3, SNR IMAGE ANALYSIS.

**2-6 PERIODIC QUALITY ASSURANCE CHECK**

On a periodic basis, such as during planned maintenance, perform the quality assurance check outlined below to ensure that the Quad T/L Spine coil is operating properly with no appreciable degradation of image quality.

1. Check the external cable for cracks or breaks once each week. See Section 4-5, CHECKING THE EXTERNAL CABLE.
2. Perform a Quad T/L Spine Coil performance verification. Refer to Section 3-2, QUAD T/L SPINE COIL IMAGING PERFORMANCE VERIFICATION.
3. Record the date and value calculated in SNR QA DATA TABLE
4. As instructed in the Data Table, divide the SNR value obtained in the periodic QA check by the original SNR value.
5. If this ratio is not greater than 85%, then there may be a problem with the Quad T/L Spine Coil or the MR system. Contact the GE Service Representative.

TABLE 2-3  
SCAN PARAMETERS

|                     |                        |                       |                 |
|---------------------|------------------------|-----------------------|-----------------|
| ID:                 | <b>geservice</b>       | Auto CF:              | <b>[Peak]</b>   |
| Name:               |                        | Receive Bandwidth:    | <b>[16kHz]</b>  |
| Weight (lb):        | <b>111</b>             | Field of View:        | <b>[24 cm]</b>  |
| Patient Entry:      | <b>[Head First]</b>    | Slice Thickness:      | <b>[10 mm]</b>  |
| Patient Position:   | <b>[Supine]</b>        | Interscan Spacing:    | <b>[0]</b>      |
| Axial/Sag Landmark: | <b>[Sternal Notch]</b> | Start Location (I/S): | <b>0</b>        |
| Coil Type:          | <b>[++] See Note</b>   | End Location (I/S):   | <b>0</b>        |
| Scan Plane:         | <b>[Axial]</b>         | No of Scan Locations: | <b>1</b>        |
| Image mode:         | <b>[2D]</b>            | FOV Center (L/R):     | <b>0(P/A):0</b> |
|                     | <b>[Monitor SAR]</b>   | Acq. Matrix: (freq.)  | <b>[256]</b>    |
| Pulse Sequence:     | <b>[Spin Echo]</b>     | Acq. Matrix: (phase)  | <b>[128]</b>    |
| Imaging Options:    | <b>[None]</b>          | Frequency Direction:  | <b>[R/L]</b>    |
| Number of Echos:    | <b>[*] See Note</b>    | Phase FOV:            | <b>default</b>  |
| Echo Time (TE):     | <b>[20 ms]</b>         | Imaging Time:         | <b>[1 Nex]</b>  |
| Rep Time (TR):      | <b>[Other] 600 ms</b>  | Contrast:             | <b>[No]</b>     |
|                     |                        | Table Delta           | <b>0</b>        |

**NOTE:** \* **[4]** for Section 3-1-2; 3-2-2  
**[1]** for Section 3-1-4; 3-2-4  
 ++ **[Body]** for Section 3-1-2; 3-2-2  
**[Other][quad t/l]** for Section 3-1-4; 3-2-4

## SECTION 3 - FUNCTIONAL CHECKS

### 3-1 SIGNA ADVANTAGE™ 1.0T SCANNER VERIFICATION

#### Note

An alternative proprietary procedure is available for GE use and to customers with a valid Advanced Service Package Limited License. Refer to "TLT PROCEDURE" located in Direction 15491, *Signa Advantage 1.5T, 1.0T, and 0.5T System Troubleshooting Guide*.

1. Scanner performance in the body coil mode should be verified. The RF power level calibration from the body coil mode is used to verify the surface coil decoupling. Landmark the 100 mm TLT phantom without the coil. Place the TLT Phantom on the Universal Phantom Holder. See ILLUSTRATION 2-1.
2. Set up the scan using the parameters in TABLE 3-1. Coil Type: **[Body Coil]** and Number of Echos: **[4]** should be inserted. Perform an **Auto Prescan** to calibrate the RF power level. Record the value for TG, R1, and R2.
3. Run the scan. Observe the resulting images. Ensure that there are no artifacts of any sort on any of the images.
4. Select **[Cancel]**, **[New Series]**, to perform SNR comparisons. Acquire two identical images using the previously loaded scan parameters in TABLE 3-1 after inserting Number of Echos: **[1]**. Immediately after acquiring the first image, select **Scan** to obtain a second. Note the series number for future reference.

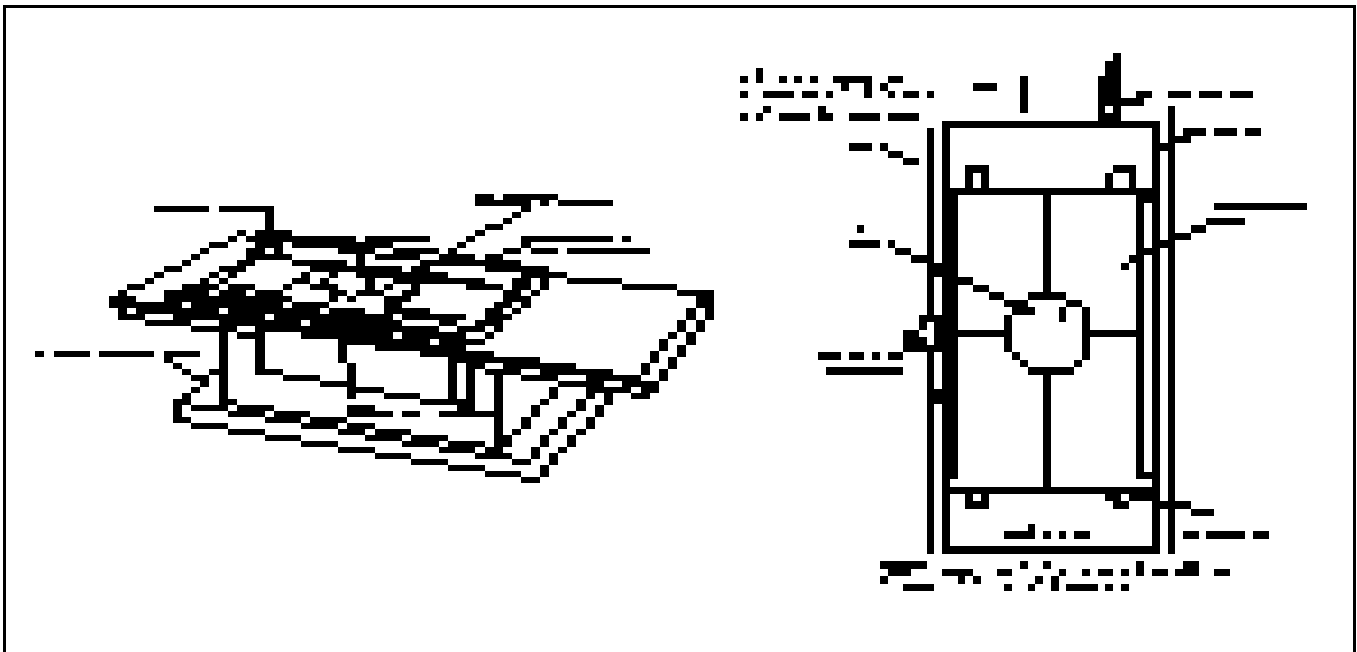
TABLE 3-1  
SCAN PARAMETERS

|                     |                        |                       |                   |
|---------------------|------------------------|-----------------------|-------------------|
| ID:                 | <b>geservice</b>       | Auto CF:              | <b>[Peak]</b>     |
| Name:               |                        | Receive Bandwidth:    | <b>[16kHz]</b>    |
| Weight (lb):        | <b>111</b>             | Field of View:        | <b>[24 cm]</b>    |
| Patient Entry:      | <b>[Head First]</b>    | Slice Thickness:      | <b>[10 mm]</b>    |
| Patient Position:   | <b>[Supine]</b>        | Interscan Spacing:    | <b>[0]</b>        |
| Axial/Sag Landmark: | <b>[Sternal Notch]</b> | Start Location (I/S): | <b>0</b>          |
| Coil Type:          | <b>[++] See Note</b>   | End Location (I/S):   | <b>0</b>          |
| Scan Plane:         | <b>[Axial]</b>         | No of Scan Locations: | <b>1</b>          |
| Image mode:         | <b>[2D]</b>            | FOV Center (L/R):     | <b>0 (P/A): 0</b> |
|                     | <b>[Monitor SAR]</b>   | Acq. Matrix: (freq.)  | <b>[256]</b>      |
| Pulse Sequence:     | <b>[Spin Echo]</b>     | Acq. Matrix: (phase)  | <b>[128]</b>      |
| Imaging Options:    | <b>[None]</b>          | Frequency Direction:  | <b>[R/L]</b>      |
| Number of Echos:    | <b>[*] See Note</b>    | Phase FOV:            | <b>default</b>    |
| Echo Time (TE):     | <b>[20 ms]</b>         | Imaging Time:         | <b>[1 Nex]</b>    |
| Rep Time (TR):      | <b>[Other] 600 ms</b>  | Contrast:             | <b>[No]</b>       |
|                     |                        | Table Delta           | <b>0</b>          |

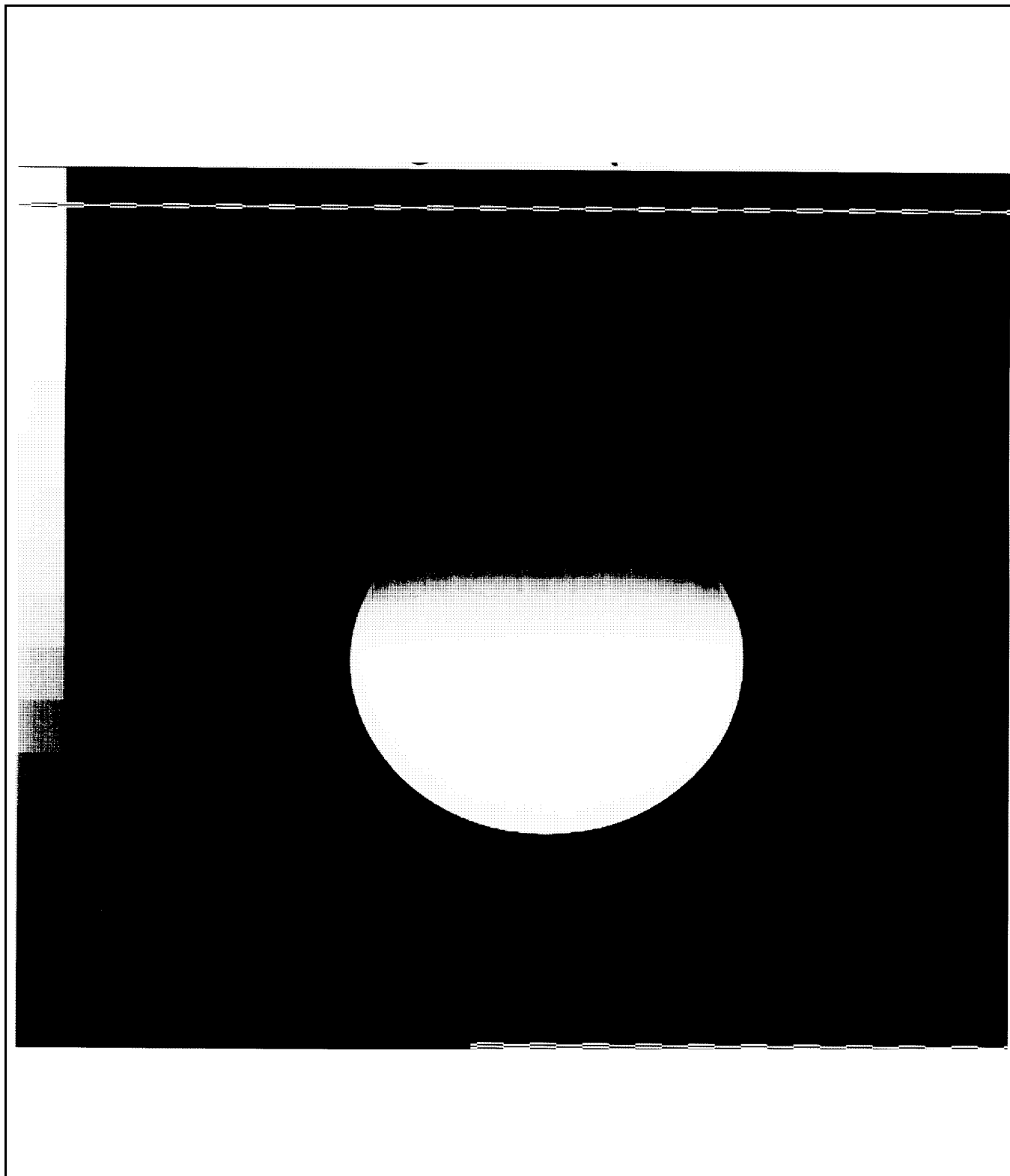
**NOTE:** \* **[4]** for Section 3-1-2; 3-2-2  
**[1]** for Section 3-1-4; 3-2-4  
 ++ **[Body]** for Section 3-1-2; 3-2-2  
**[Other][quad t/l]** for Section 3-1-4; 3-2-4

### 3-2 QUAD T/L SPINE COIL IMAGING PERFORMANCE VERIFICATION

1. Set up the Quad T/L Spine coil and 100 mm phantom as shown in ILLUSTRATION 3-1. Connect the surface coil cable to the **SURFACE COIL QUICK DISCONNECT BOX** and connect to the **HEAD COIL PORT**.
2. Select **[Cancel]**, **[New Series]**, and repeat the scan using the parameters in TABLE 3-1. Coil Type: **[Other][quad t/l]** and Number of Echos: **[4]** should be used. Use **Auto Prescan** to set the values of R1 and R2 to those in Section 3. If the value of TG does not match the value obtained in step 3-1-2, use **Manual Prescan** to set the value of TG.
3. Run the scan. Ensure that there are no artifacts of any sort on any of the images. Study the image using a **Window Width** and **Window Level** to obtain good contrast. A properly functioning coil will produce an image with a smooth signal pattern, gradually dropping in intensity as the distance from the coil increases, see ILLUSTRATION 3-2. A decoupling failure will cause holes, distortions, or striped patterns in the image, Refer to Section 3-4 to 3-7 to troubleshoot decoupling failures.
4. Repeat step 2 with Number of Echos: **[1]** inserted. Run the scan. Immediately after acquiring the first image, select **Scan** to obtain a second.
5. Compute the SNR using the procedure described in Section 3-3, SNR IMAGE ANALYSIS. Compare the signal to noise ratio of the first echo of the Quad T/L Spine Coil images with the signal to noise ratio of the first echo of the body coil images of step 3-1-4. The Quad T/L Spine Coil scans should display a signal to noise ratio greater than 2.5 times that of the body coil scans.
6. Inspect the images for visible ghosting (similar in appearance to motion artifacts) in the phase encoding direction. If images contain ghosting artifacts, suspect an intermittent cable or connection. Refer to Section 3-4 to 3-7 for troubleshooting information.



PHANTOM PLACEMENT  
ILLUSTRATION 3-1



NORMAL DECOUPLING  
ILLUSTRATION 3-2

### 3-3 SNR IMAGE ANALYSIS

#### Description

The SNR tool retrieves two operator selected images. Signal value is computed as the mean pixel value in a ROI covering 80% of the image. The image is analyzed to determine the center for positioning the ROI. A difference image is created by subtracting the second image from the first and to calculate noise from the subtracted image. Signal value, noise value, and SNR are reported. The difference image can be saved with the results annotated (option).

#### Procedure

1. Touch **[Utilities]**, **[MR Tools]**, **[Image Quality]**, then **[SNR Test]**. The SNR Test screen is displayed. See ILLUSTRATION 3-3.
2. Enter first image exam, series, and image numbers. If exam, series, or image numbers are not known, select **[List Exams]**, **[List Series]**, or **[List Images]** to display list to choose from. Enter second image series and image number. The second image Exam Number is defaulted to the same as first image.

#### Note

Image number selection must be back lit (highlighted) to be able to enter information. Use Switch key on keyboard to transfer control from left to right side of Touch Screen.

3. Select **[Filter Off]** and **[Image Off]**. Touch **[Accept]** to begin analysis. The final values are displayed on the screen.
4. Record Signal to Noise Ratio on SNR QA DATA TABLE. Touch **[Continue]**, select the next exam and repeat for each pair.

| NEMA STANDARD SNR TEST                                                                                                                 |                   |                    |                    |  |
|----------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------|--------------------|--|
|                                                                                                                                        | Filter<br>Off     | Image<br>Off       |                    |  |
| First<br>Image                                                                                                                         | List/Sel<br>Exams | List/Sel<br>Series | List/Sel<br>Images |  |
| Second<br>Image                                                                                                                        |                   | List/Sel<br>Series | List/Sel<br>Images |  |
| <div> <div>Cancel</div> <div>Utility<br/>Main</div> <div>MR Tools<br/>Main</div> <div>Image<br/>Quality</div> <div>Accept</div> </div> |                   |                    |                    |  |



Note: Accept changes to continue only after an analysis has been performed.

SNR TEST SCREEN  
ILLUSTRATION 3-3



### 3-4 COIL IMPEDANCE MEASUREMENTS

1. Move to an area where the magnetic field from the Signa Advantage 1.0T magnet is less than 5 Gauss. Load the surface coil with a human subject. Use the procedure in the Operation Manual to place the surface coil.
2. Connect the surface coil cable to the vector impedance meter probe using an appropriate adapter. Use the vector impedance meter to measure the coil impedance at 42.68 MHz. Ensure that the cable does not touch or pass close to the coil housing while making this measurement.
3. The impedance magnitude should nominally be 50 ohms under loaded conditions. It should fall within the range of 37.5 to 62.5 ohms under test conditions.
4. The phase angle should nominally be 0 degrees. It should fall within the range of -17 to +17 degrees under test conditions.

### 3-5 COIL INTEGRITY TESTS

1. Repeat Section 3-4 while flexing the surface coil cable, especially at the fitting and the coil housing strain relief. Using the **HIGH SPEED** mode on the vector impedance meter, note any erratic fluctuations in the coil impedance. Any erratic or unstable readings may indicate a defective cable or connector. Refer to Section 4, REPLACEMENT/MAINTENANCE.

### 3-6 PROTECTION DIODE TESTS

1. Use the digital multimeter on the diode test range to measure across the coil connector. With the negative lead to the connector shell and the positive to the connector center pin, the reading should be between 0.4 and 0.9. Reversing the leads should cause the reading to be infinity (overrange).
2. An open reading in both directions indicates an open diode, or an open cable. Refer to Section 3-8 to check the cable. Refer to Section 4-3, DIODE REPLACEMENT to replace the diode.
3. A short circuit or excessively low reading indicates a shorted diode or a shorted cable. Refer to Section 3-8 to check the cable. Refer to Section 4-7, DIODE REPLACEMENT to replace the diode.

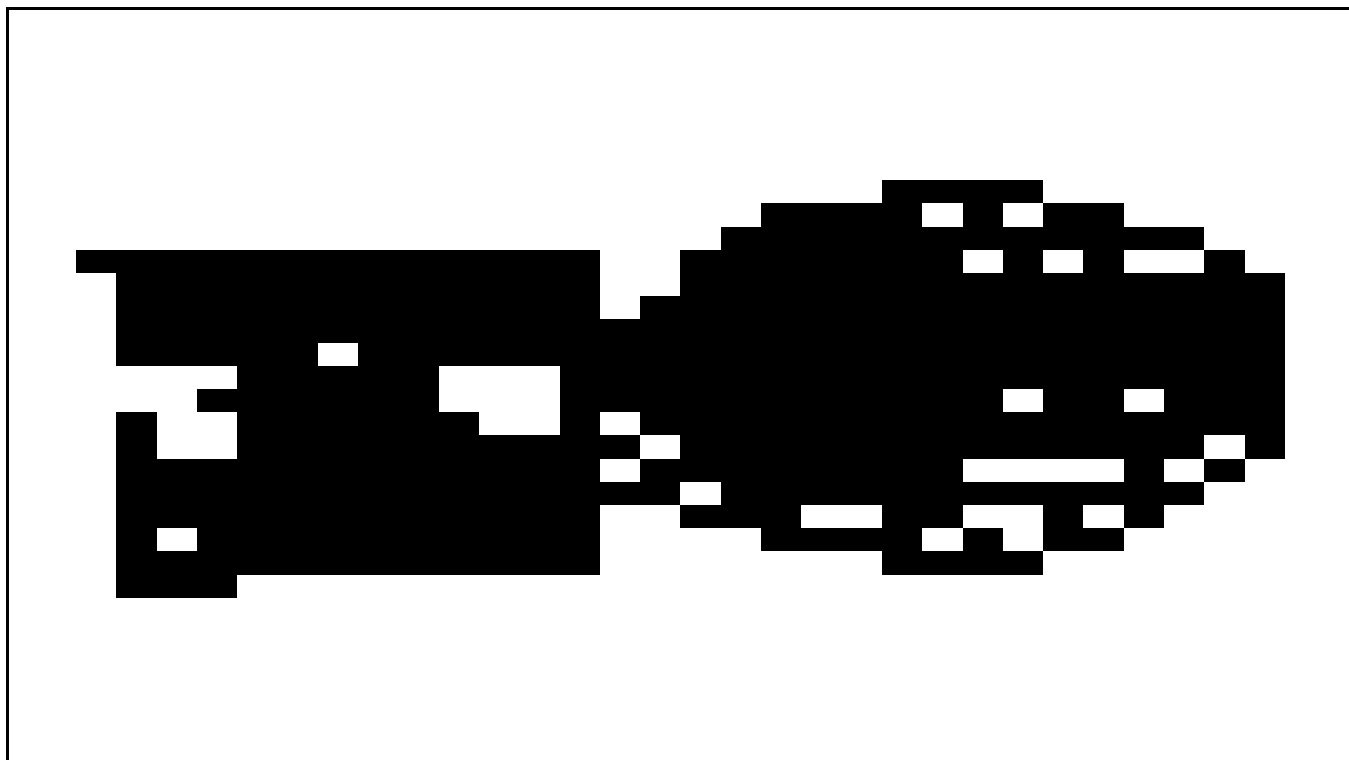
**3-7 DECOUPLING DIODE TESTS****WARNING**

**DO NOT MOVE THE BALUN BOARD COMPONENTS WHEN MAKING THESE MEASUREMENTS. ANY MOVEMENT OF THE COMPONENTS MAY DETUNE THE COIL CIRCUITRY.**

**Note**

Decoupling diodes are not field replaceable. Coils with decoupling diode failures must be replaced with a new coil assembly.

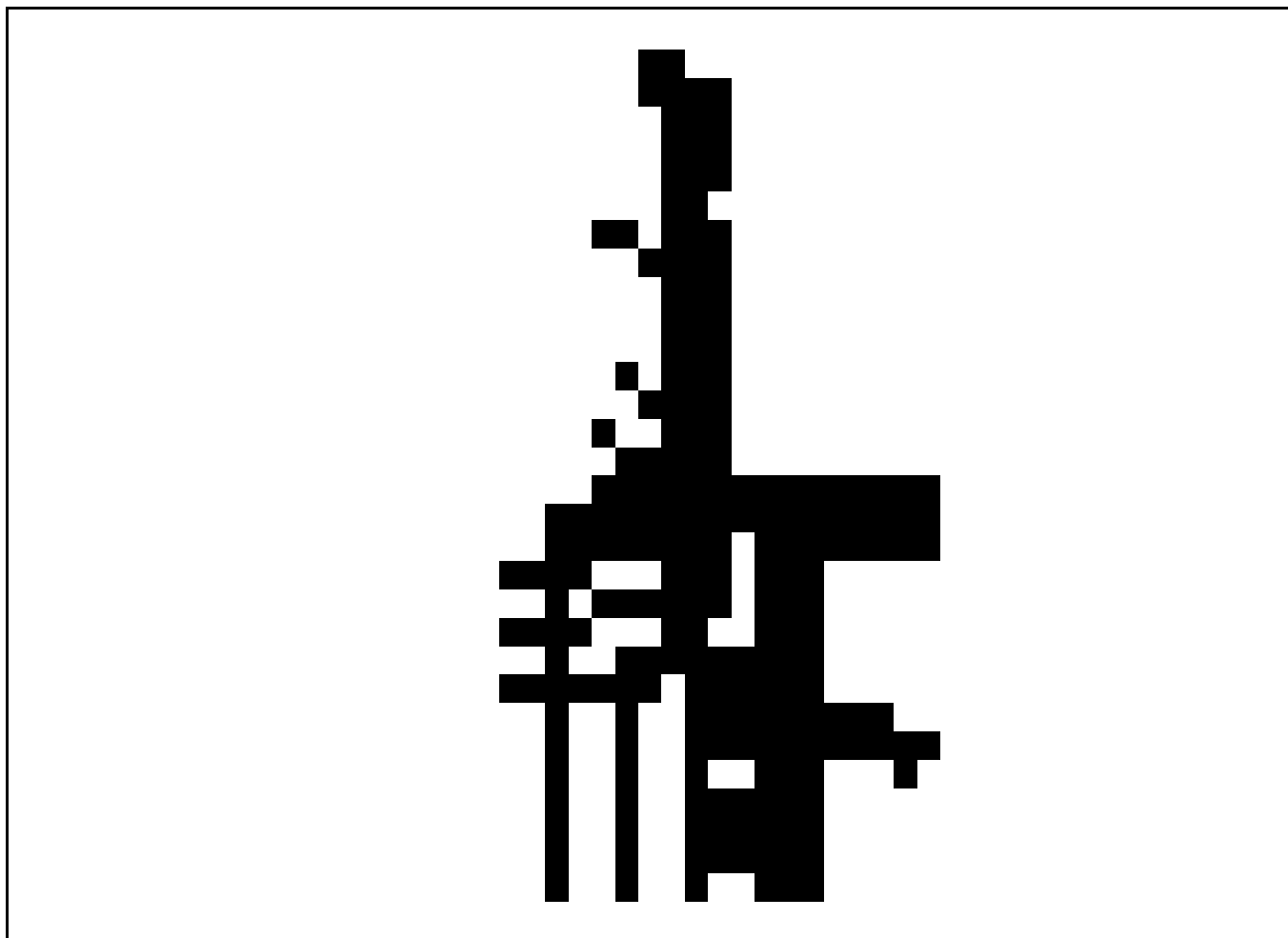
1. Remove the service access cover.
2. Locate the three pairs of test points for the decoupling diodes labeled V, H1, and H2 in ILLUSTRATION 3-4. Using the diode test function on the digital multimeter, measure across each pair of points in sequence, first across the pair marked **V**. A reading between 0.5 and 1.0 should be obtained. Reverse the leads on the same test points to reverse the polarity of the multimeter. A similar reading should be obtained.
4. An overrange in both directions indicates that both diodes of a pair are open. This condition causes a portion of the decoupling circuitry to be inoperative, and the coil must not be used clinically.
5. A reading near zero indicates one or more shorted diodes. While decoupling is not affected, the signal to noise ratio of the coil will be degraded.



**COIL TEST POINTS**  
ILLUSTRATION 3-4

**3-8 CABLE TESTS**

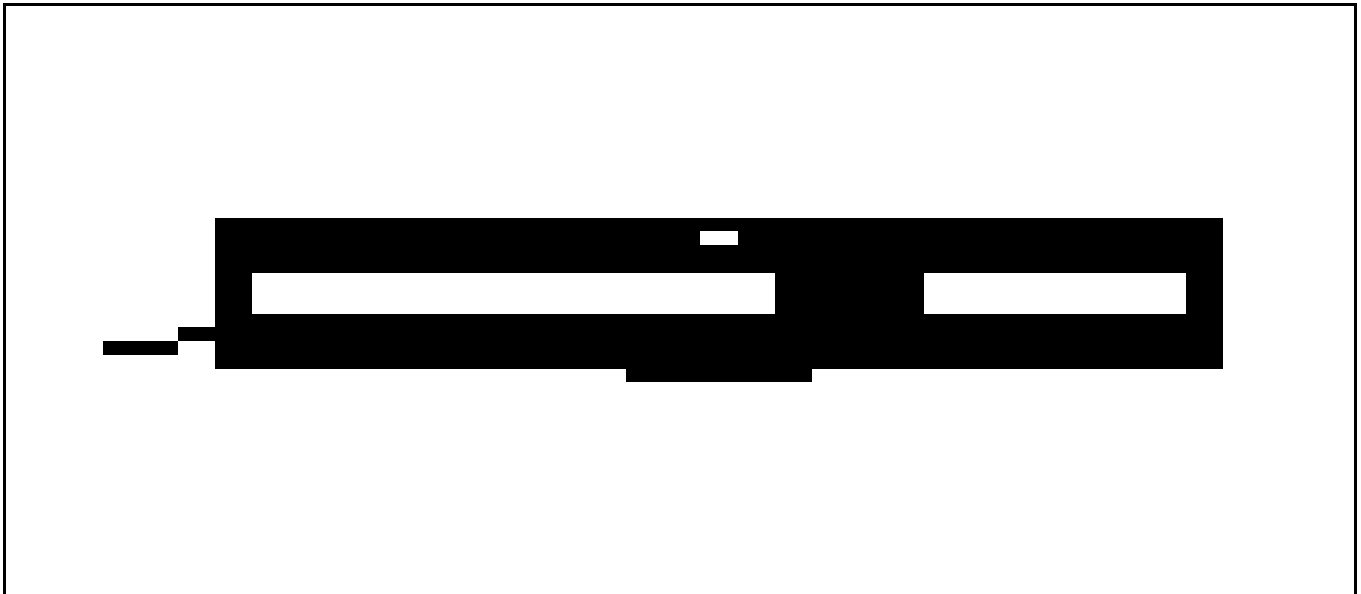
1. If the surface coil passes the tests in Sections 3-4, 3-5, and 3-6, the cable can be assumed to be functional.
2. If a short or open circuit was discovered in Section 3-4, the cable may be tested by disconnecting the cable at the PC board via the service access cover, as shown in ILLUSTRATION 3-5. The cable may then be tested for continuity of the center conductor and shield, and for shorts between the center conductor and shield using the multimeter continuity test function.



**EXTERNAL CABLE CONNECTION**  
ILLUSTRATION 3-5

**3-9 VERIFICATION OF A DRIVE CORD PROBLEM**

1. Turn the housing over, allowing it to rest on the top cover. See ILLUSTRATION 3-6.
2. Slide the Positioning Handle to the end opposite the cable exit.
3. While looking through the clear service access cover, slide the handle back and forth and verify that the handle position is within +/- 0.25 inch of the actual coil position.
4. If the handle moves more than +/- 0.25 inch prior to coil movement in either direction, one or both of the drive cords have stretched beyond acceptable limits. If the coil does not move at all, one or both of the drive cords are broken. See Section 4-4, DRIVE CORD ADJUSTMENT AND REPLACEMENT.

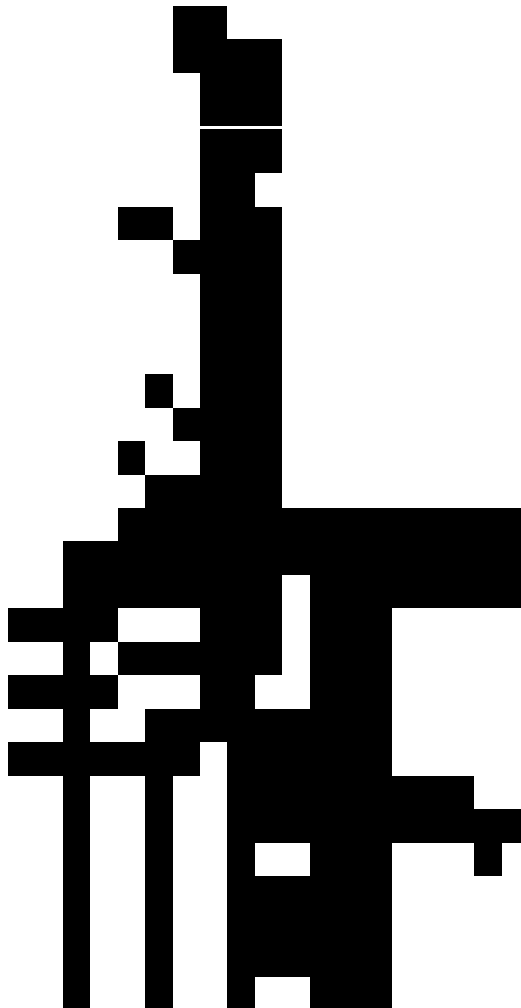


EXTERNAL CABLE CONNECTION  
ILLUSTRATION 3-6

## SECTION 4 - REPLACEMENT/MAINTENANCE

### 4-1 EXTERNAL COAXIAL CABLE REPLACEMENT

1. Loosen and remove the strain relief body from the strain relief base. Slide the strain relief body down the cable away from the base. See ILLUSTRATION 4-1.
2. Loosen and remove the retaining nut from the strain relief base.
3. Remove the cable SMA connector from the SMA feedthrough connector on the bottom cover of the housing.
4. Remove the strain relief base from the threaded hole in the top cover of the positioner.



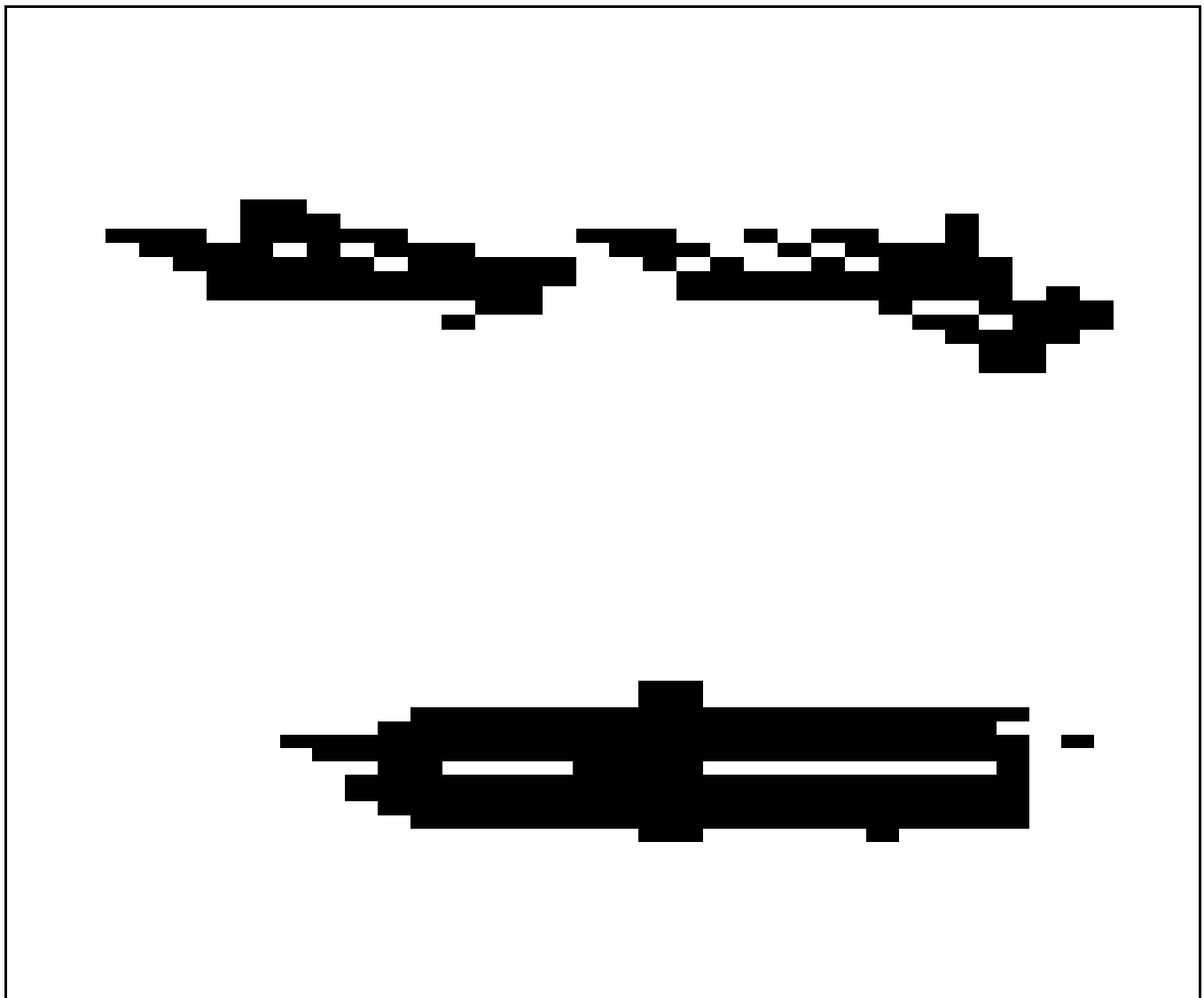
COAXIAL CABLE REPLACEMENT  
ILLUSTRATION 4-1

**4-1 EXTERNAL COAXIAL CABLE REPLACEMENT - continued**

5. Loosen and remove the new strain relief body from the new strain relief base. Remove the retaining nut from the new external cable assembly. Insert the SMA end of the cable through the strain relief mounting hole in the positioner top cover. Place the retaining nut over the SMA end of the cable.
6. Connect the cable SMA connector to the feedthrough connector on the positioner bottom cover and tighten firmly.
7. Attach the strain relief base to the top cover of the positioner housing. Tighten one-quarter turn beyond initial contact. Install and tighten the retaining nut. Tighten the nut one-half turn beyond initial contact with the surface of the top cover.
8. Install and tighten the strain relief body until the base begins to turn [8 inch-pounds].
9. Repeat the tests in Sections 3-1, 3-2, and 3-3 to verify proper coil operation.

**4-2 POSITIONER DISASSEMBLY INSTRUCTIONS**

1. Remove the comfort pad from the top of the coil housing; turn the coil housing over to expose the bottom side. See ILLUSTRATION 4-2.
2. Remove the clear service access cover.
3. Disconnect the internal cable from the balun board by unscrewing the SMA connector. Note the polarity setting for assembly of the coil.
4. Remove the external cable assembly as described in Section 4-1
5. Remove the nylon screws located beneath the coil housing on the positioning handle side of the coil housing.



POSITIONER ASSEMBLY/DISASSEMBLY  
ILLUSTRATION 4-2

**4-2 POSITIONER DISASSEMBLY INSTRUCTIONS - continued**

6. Remove the carrying handle by removing the two knurled nuts which retain the handle, and the two springs.
7. Remove the nine nylon drive rivets [two on each end, five on the carrying handle side of the coil housing] by punching the pin into the coil housing using a 3/32 inch transfer punch and a small hammer. Pull the rivet body out of the hole after the pin has been driven into the coil housing. See top of ILLUSTRATION 4-2.
8. Remove the ten brass nuts and washers located on the bottom cover of the coil housing.
9. Rest the coil housing on its bottom cover, and move the coil to the center of its travel. Carefully spread the slot sufficiently to push the positioning handle through the slot into the coil housing. Lift off the top cover.
10. Locate and remove the four drive rivet pins previously pushed into the coil housing.



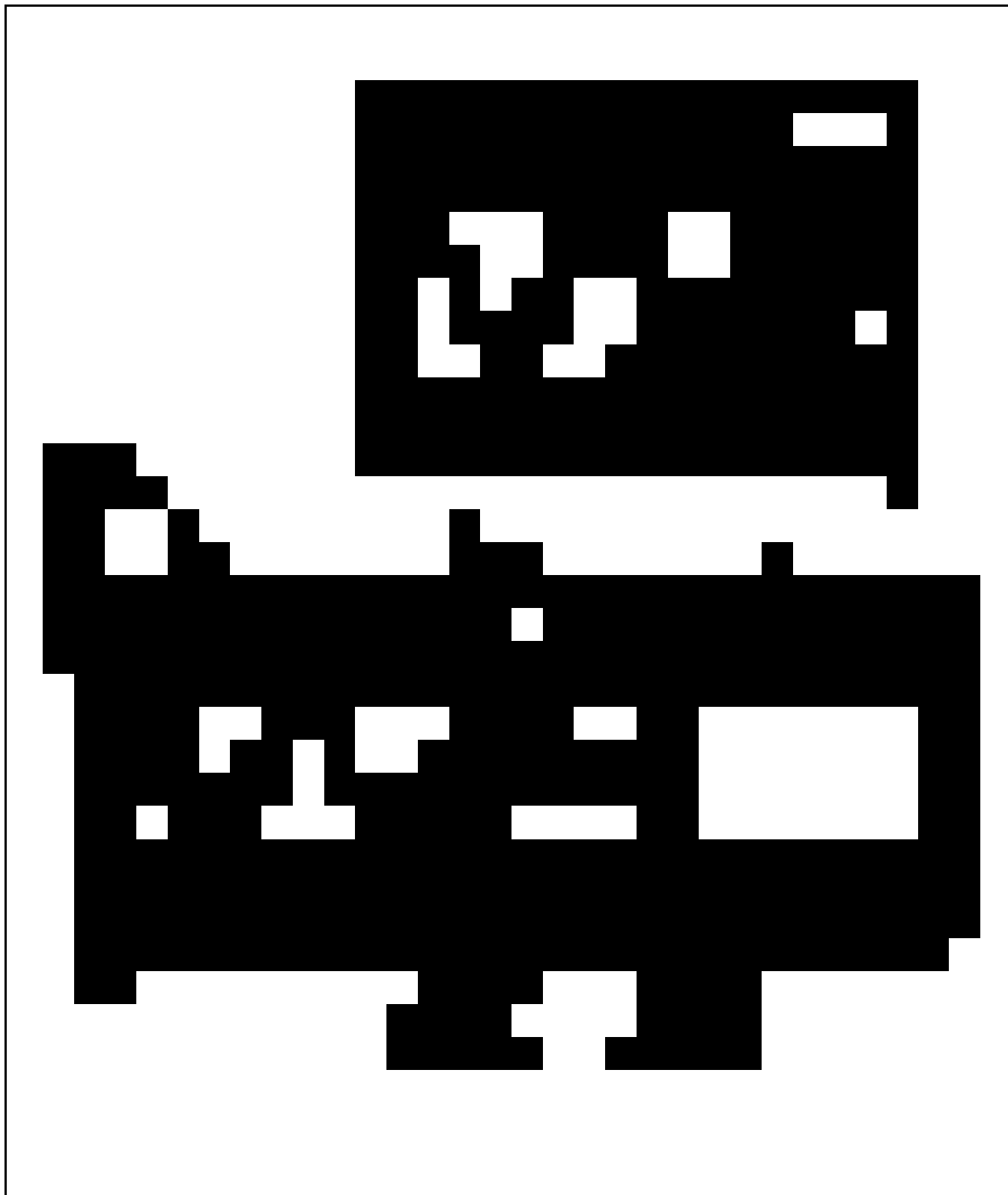
**4-3 INTERNAL COAXIAL CABLE REPLACEMENT**

1. Remove the external coaxial cable as described in Section 4-1.
2. Remove the retaining nut and washer from the SMA feedthrough connector in the bottom cover. See ILLUSTRATION 4-1.
3. Disassemble the unit as described in Section 4-2.
4. Remove the binding posts, items 1 and 2 in ILLUSTRATION 4-3.
5. Remove the cable clamp, item 3.
6. Remove the SMA feedthrough connector from the bottom cover.
7. Withdraw the internal coaxial cable from the cable management system.
8. Feed the new cable in the same path as the old. Push the SMA feedthrough connector through the hole in the bottom cover and secure with the nut and washer to 10 inch-pounds.
9. Place the cable clamp (item 3) on the cable 5 1/4 inches from the SMA fitting. Orient the clamp as shown in the illustration. Tighten to 3 inch-pounds.
10. Install new binding posts and screws (items 1 and 2) to reconnect the cable management system to the Coil assembly and the mounting bracket. Push the parts together. Do not thread the screw into the post.

**NOTE**

The cable management linkage bars must be attached to the bottom side of the Coil assembly and the mounting bracket.

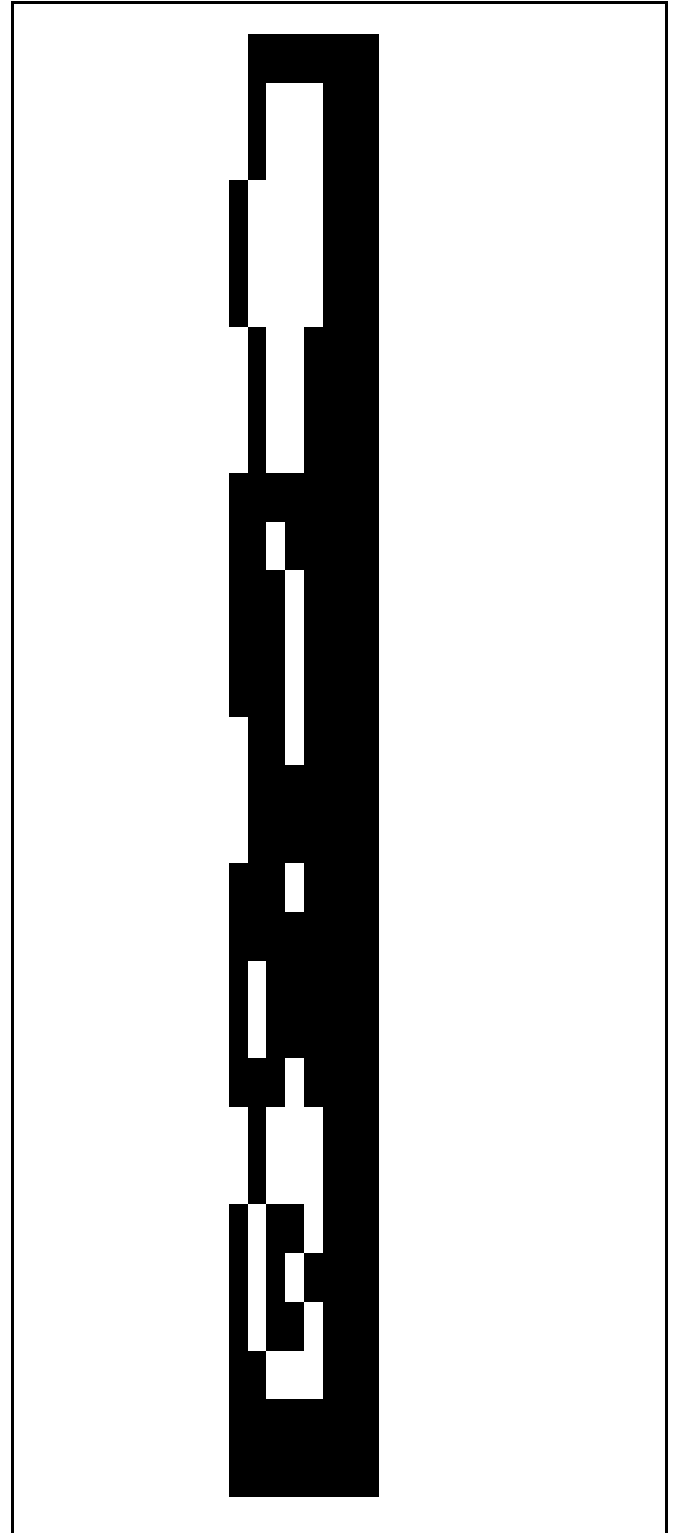
11. Verify the operation of the drive system by sliding the connector block several times from one end of its travel to the other.
12. Assemble the coil housing following the steps in Section 4-5.
13. Repeat the tests in Sections 3-1, 3-2, and 3-3 to verify proper coil operation.



INTERNAL COAXIAL CABLE REPLACEMENT  
ILLUSTRATION 4-3

#### 4-4 DRIVE CORD ADJUSTMENT AND REPLACEMENT

1. Disassemble the unit as described in Section 4-2.
2. To tighten the drive cord, loosen the lock nuts [items 4] and tighten the tension adjuster bolts [items 5] to obtain alignment between the coil and the indicator. Tighten the lock nuts one-quarter turn beyond contact with the indicator block. See ILLUSTRATION 4-3.
3. To replace the drive cord, remove the lock nuts [items 4] and the tension adjuster bolts [items 5]. Remove the cord from the coil assembly.
4. Place the new cord in the appropriate holes in the coil assembly from the top. Tie a double knot in the cord underneath the coil assembly and cut off the excess cord. See ILLUSTRATIONS 4-3, 4-4.
5. Route the cord around the inner posts closest to the coil assembly and crisscross the two cords. Route the cord around the outer posts on the opposite sides of the coil housing.
6. Feed the opposite end of the cord through the tension adjuster bolts and tie a double knot to achieve the required finish length. Cut off the excess cord.
7. Replace the tension adjuster bolts in the indicator block. Adjust the bolts as described in step 4-4-2.
8. Verify the operation of the drive system as described in step 4-3-11.
9. Assemble the coil housing as described in Section 4-5.
10. Repeat the tests in Sections 3-1, 3-2, and 3-3 to verify proper coil operation.



DRIVE CORD ADJUSTMENT AND REPLACEMENT  
ILLUSTRATION 4-4

**4-5 ASSEMBLY OF THE COIL AND POSITIONER**

1. With the coil assembly set to the center of its travel, place the top cover on the bottom cover. Spread the slot in the top cover sufficiently to pull the positioning handle through. Press the top cover down on the bottom cover, making certain the ten brass studs protrude through the holes in the bottom cover. See ILLUSTRATION 4-2
2. While holding the top and bottom covers securely together, turn the coil over on the top cover. Install the ten brass washers and nuts on the ten studs. Torque to 20 inch-pounds. Slide the coil from one end of travel to the other to ensure that there is no binding of the coil positioning hardware.
3. Insert nine new drive rivets supplied with the service kit into the appropriate holes in the positioner housing. Using a small hammer, drive the center pins into the rivet bodies until flush.
4. Attach the carrying handle using the two knurled nuts and two springs.
5. Replace the nine nylon screws located beneath the scale. Torque to 2 inch-pounds.
6. Install the external cable as described in Section 4-1.
7. Connect the internal coaxial cable to the SMA fitting on the balun board.
8. Attach the clear service access cover using the six nylon screws. Torque to 1 inch-pound.
9. Return the coil housing to its normal position and attach the comfort pad.
10. Repeat the tests in Sections 3-1, 3-2, and 3-3 to verify proper coil operation.

**4-6 OUTPUT CONNECTOR/CABLE  
REPLACEMENT**

1. If the cable will be shortened to less than 71 inches by a repair, it is recommended that the entire cable be replaced in the event of a connector or connector-end cable failure.
2. Replace the connector using ILLUSTRATION 4-5 as a guide. Use a new BNC coaxial fitting - part number 46-271494P1, and crimp tool - part number 46-255841, with insert - part number 46-255841P100, to install the replacement connector on the cable.
3. Repeat the tests in Sections 3-1, 3-2, and 3-3 to verify proper coil operation.

**4-7 DIODE REPLACEMENT**

1. As coil tuning, orthogonality, and decoupling may be affected by decoupling diode replacement, decoupling diodes are not field replaceable. Coils with decoupling diode failure must be replaced.
2. The protection diodes are in parallel. One of the diodes is inaccessible and cannot be changed. Coils with protection diode failure must be replaced.

**4-8 REPLACING THE MECHANICAL HARDWARE**

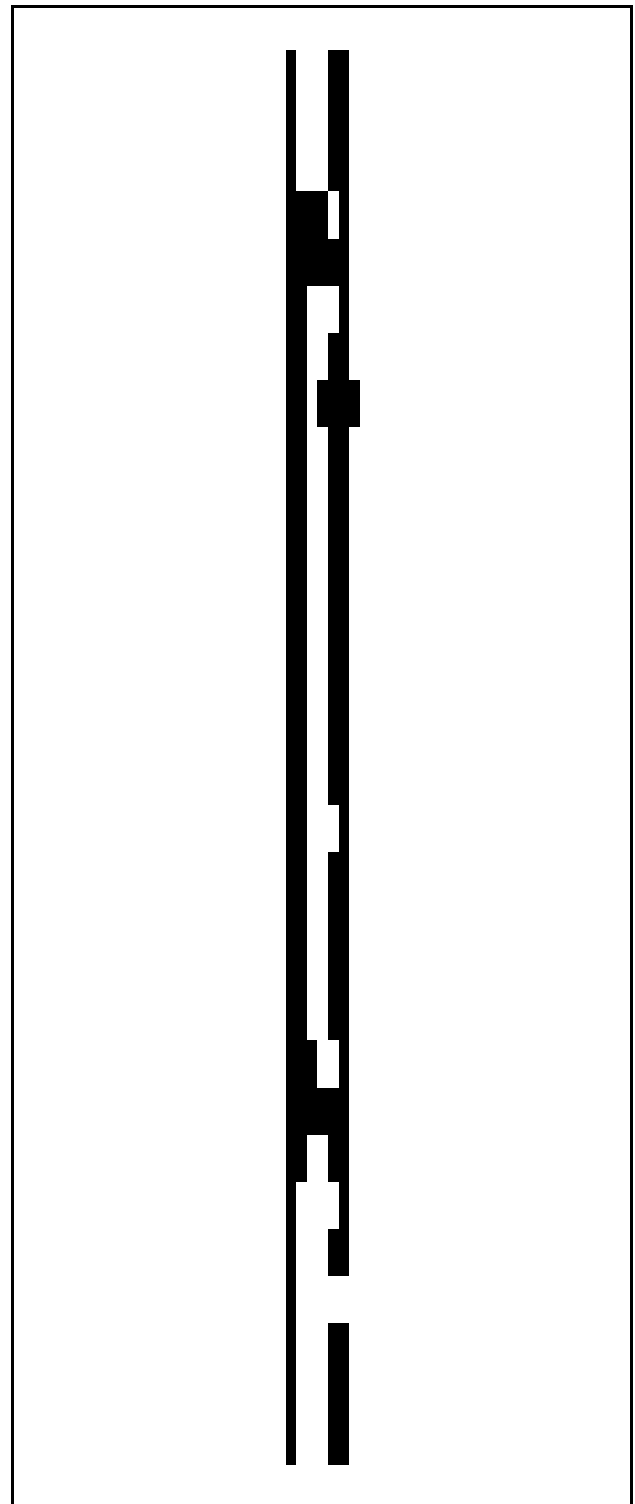
1. Replace the entire Quad T/L Spine Coil. Refer to Section 5, RENEWAL PARTS, for Quad T/L Spine Coil part number.

**4-9 CHECKING THE EXTERNAL CABLE**

1. Check the external cable for cracks or breaks once each week. Replace the external cable if any damage or wear is found. Refer to Section 4-2, OUTPUT CONNECTOR REPLACEMENT, for replacement procedure.

**4-10 CLEANING THE QUAD T/L SPINE COIL**

1. Clean the Quad T/L Spine Coil and external cable with a mild dishwasher liquid and water solution. Wet a soft cloth with the solution and proceed to clean.



**OUTPUT CONNECTOR/CABLE REPLACEMENT**  
ILLUSTRATION 4-5

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DIRECTION 2100624

## SECTION 5 - RENEWAL PARTS

### 5-1 SERVICE KITS

1. A service kit is available for replacement of the cable assembly:

**POSITIONER REPAIR KIT 46-317047P1** — Consisting of the following items:

| ITEM | DESCRIPTION             | QTY    |
|------|-------------------------|--------|
| 1    | Drive Cord              | 12 ft. |
| 2    | Drive Rivets            | 9      |
| 3    | Binding Post Fastener   | 2      |
| 4    | 4-40 Nylon Fastener     | 2      |
| 5    | Black Indicator Cap     | 2      |
| 6    | Internal Cable Assembly | 1      |

2. The Quad T/L Spine Coil Part number is:

| ITEM | DESCRIPTION                                | QTY |
|------|--------------------------------------------|-----|
| 1    | Quad T/L Spine Coil with Cable—46-328295P1 | 1   |

3. The External Cable Part number is:

| ITEM | DESCRIPTION                  | QTY |
|------|------------------------------|-----|
| 1    | External Cable Kit—2100937-4 | 1   |

Blank

DIRECTION 2100624



## SNR QA DATA TABLE

Use the space provided below to record the calculated signal to noise ratio—SNR, data obtained from Section 4, *SNR IMAGE ANALYSIS*. After recording the SNR data obtained during the initial coil installation, record subsequent SNR data in column 2, enter the date in column 1 as a periodic QA check. If the ratio of any of the coils found is not greater than 85%, then there is a problem in the coil or the MR system.

Original SNR data obtained at initial coil installation:

Body Coil SNR Value: \_\_\_\_\_

Quad T/L Spine SNR value: \_\_\_\_\_

Date: \_\_\_\_\_

### SNR DATA QA (QUALITY ASSURANCE) CHECK:

| 1     | 2                                          | 3                                                  |     |
|-------|--------------------------------------------|----------------------------------------------------|-----|
| Date  | Quad T/L<br>Spine Coil<br>QA SNR<br>Value: | Is new value divided<br>by original value<br>>85%? |     |
| _____ | _____                                      | _____                                              | Y/N |
| _____ | _____                                      | _____                                              | Y/N |
| _____ | _____                                      | _____                                              | Y/N |
| _____ | _____                                      | _____                                              | Y/N |
| _____ | _____                                      | _____                                              | Y/N |
| _____ | _____                                      | _____                                              | Y/N |
| _____ | _____                                      | _____                                              | Y/N |
| _____ | _____                                      | _____                                              | Y/N |
| _____ | _____                                      | _____                                              | Y/N |
| _____ | _____                                      | _____                                              | Y/N |
| _____ | _____                                      | _____                                              | Y/N |

Make additional copies of this document as needed.



## DEFECTIVE SURFACE COIL RETURN FORM

### NOTE

For proper assessment of defective returned coils, both sides of this form should be completely filled out and accompany all returned coils. Include films or prints of any image quality complaints with a description of the scan prescription used.

DATE: \_\_\_\_\_

SITE NAME: \_\_\_\_\_

SITE ADDRESS: \_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

SERVICE ENGINEER: \_\_\_\_\_

COIL SERIAL NUMBER: \_\_\_\_\_

DATE COIL INSTALLED: \_\_\_\_\_

DESCRIPTION OF COIL PROBLEM: \_\_\_\_\_  
[Please be descriptive;"Broken" is not enough] \_\_\_\_\_

| ELECTRICAL CHECKS                                                                                  |  |
|----------------------------------------------------------------------------------------------------|--|
| VECTOR IMPEDANCE METER TEST - COIL LOADED WITH HUMAN SUBJECT IN FREE SPACE—Refer to Section 3-4-1. |  |
| MAGNITUDE                                                                                          |  |
| PHASE                                                                                              |  |
| VECTOR IMPEDANCE METER TEST - COIL UNLOADED IN FREE SPACE—Refer to Section 3-4-1.                  |  |
| MAGNITUDE                                                                                          |  |
| PHASE                                                                                              |  |
| PIN DIODE TEST—Refer to Section 3-4-3.                                                             |  |
| DIODE DROP—FORWARD BIAS                                                                            |  |
| DIODE DROP—REVERSE BIAS                                                                            |  |

**Note**

An alternate proprietary procedure is available to GE use and to customers with a valid Advanced Service Package Limited License. Refer to "TLT PROCEDURE" located in Direction 15491, *Signa Advantage 1.5T, 1.0T & 0.5T System Troubleshooting Guide*. Use the table to record the TLT results of the defective coil.

TABLE DR-1  
TLT DATA FOR DEFECTIVE SURFACE COIL

| SITE:            |            | NAME: | DATE: |
|------------------|------------|-------|-------|
| TLT FILE NUMBER: |            |       |       |
| SNR:             | NOISE      |       |       |
|                  | SNR MEAN   |       |       |
|                  | SNR SIGNAL |       |       |
|                  | SNR AREA   |       |       |
| TRMAP:           | 89-91      | %     | %     |
|                  | 85-95      | %     | %     |
|                  | 65-115     | %     | %     |
|                  | FLIP MEAN  |       |       |
|                  | FLIP SDV   |       |       |
|                  | RCV MEAN   |       |       |
|                  | RCV SDV    |       |       |
|                  | DIFF MEAN  |       |       |
|                  | DIFF SDV   |       |       |





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